```
> restart;#The photon mass
> gram:=5.6*10^32;#1 gram is 5.6*10^32 eV/c^2;
```

$$gram := 5.600000000 \times 10^{32} \tag{1}$$

```
> c := 299792458;
```

$$c := 299792458 \tag{2}$$

```
> gram^(-1);
```

$$1.785714286 \times 10^{-33} \tag{3}$$

```
> 10^(-9)/(5.6*10^32)
```

$$1.785714286 \times 10^{-42} \tag{4}$$

```
> restart; c := 299792458; ec := 1.602176000*10^(-19); Digits := 28;
```

$$c := 299792458$$

$$ec := 1.602176000 \times 10^{-19}$$

$$Digits := 28 \tag{5}$$

```
> ec;
```

$$1.602176000 \times 10^{-19} \tag{6}$$

```
> #photon rest mass in J of wave length 555 nm = 2.5669679·10^-38 J
    but The rest mass don't varies much with frequency
> solve( 0.00678 / (1 - V^2/c^2)^(1/2) = 0.08, V ); #The magnetron
```

$$-2.98713878735712876861526856 8 \times 10^{8}, 2.98713878735712876861526856 8 \times 10^{8} \tag{7}$$

```
> V := 2.941137856159333281609533114·10^8;
```

$$V := 2.941137856159333281609533114 \times 10^{8} \tag{8}$$

```
> 1-V/c;# The deley of photnos of 555nm with respect to c;
```

$$0.018942012157179323834246319 2 \tag{9}$$

```
> 1239.84/2 ; 1239.84/12.64
```

$$619.92000000000000000000000$$

$$98.088607594936708860759493 67 \tag{10}$$

```
> 6.78·10^-3/1239.84·10^3;
    #the shift in anngstrom of wavelenght between photons 400nm
    and 650nm thus arund ~0.0054 Angstrom that sould be detected
    from a Galaxy 100 million LY away
```

$$0.005468447541618273325590398761 \tag{11}$$

```
> #Cepheid variables must be used to measure distances smaller than
    70-100 million LY to other Galaxy thus the redshift
> #Thus the photon rest mass should be around ~ 10^-8 - 10^-9 eV
> 1239.84/400 ; 1239.84/650;
```

$$3.099600000000000000000000000$$

$$1.907446153846153846153846154 \quad (12)$$

```
> V0 := solve( 10^-12 / (1 - V0^2/c^2)^(1/2) = 3.0996, V0 );
```

$$V0 := -2.997924579999999999999999844 \times 10^8, 2.997924579999999999999999844 \times 10^8 \quad (13)$$

```
> V1 := solve( 10^-12 / (1 - V1^2/c^2)^(1/2) = 1.907446, V1 );
```

$$V1 := -2.997924579999999999999999588 \times 10^8, 2.997924579999999999999999588 \times 10^8 \quad (14)$$

```
> V0/V1; E0 := 3.0996; E1 := 1.907446;
```

$$\frac{-2.997924579999999999999999844 \times 10^8, 2.997924579999999999999999844 \times 10^8}{-2.997924579999999999999999588 \times 10^8, 2.997924579999999999999999588 \times 10^8}$$

$$E0 := 3.0996$$

$$E1 := 1.907446 \quad (15)$$

```
> V0 := (1 - (3·10^-9·E0^-1)^2)^0.5·c; V0/(c·4·10^-8);
   #Difference at 17 decimal place
```

$$V0 := 2.997924579999999998595822815 \times 10^8$$

$$2.499999999999999998829042270 \times 10^7 \quad (16)$$

```
> V1 := (1 - (3·10^-9·E1^-1)^2)^0.5·c; V1/(c·4·10^-8);
   #Difference at 17 decimal place
```

$$V1 := 2.997924579999999996292094024 \times 10^8$$

$$2.499999999999999996907939245 \times 10^7 \quad (17)$$

```
> V0/c;#Difference at 18 decimal place
```

$$0.999999999999999999995316169078 \quad (18)$$

```
> V1/c; #Difference at 17 decimal place
```

$$0.999999999999999999987631756980 \quad (19)$$